(19) LE GOUVERNEMENT
DU GRAND-DUCHÉ DE LUXEMBOURG
Ministère de l'Économie

(11) N° de publication : **LU502887**

(12) **BREVET D'INVENTION** **B1**

(21) N° de dépôt: LU502887

(22) Date de dépôt: 11/10/2022

(51) Int. Cl.:
G06N 3/02, G06Q 50/18

(30) Priorité:

(43) Date de mise à disposition du public: 11/04/2023

(47) Date de délivrance: 11/04/2023

(73) Titulaire(s):
WUXI PEOPLE'S HOSPITAL – 214000 Wuxi, Jiangsu (Chine)

(72) Inventeur(s):
LU Peihua – Chine

(74) Mandataire(s):
Biopatents IP Consultancy – 5691NW Son (Pays-Bas)

(54) **Patent innovation method and system based on artificial intelligence.**

(57) The invention relates to the technical field of intellectual property rights, and in particular to a patent innovation method and system based on artificial intelligence. The patent innovation system comprises the following modules: an input module is used for inputting research objects to be innovated; a database module is used to obtain patent technical documents and information published by countries all over the world; an artificial module is used to analyze the processing database module, so as to put forward a new and innovative technical scheme; an output module is used to output the innovation results of the artificial intelligence module, thus generating a large number of innovative patents. This invention makes use of artificial intelligence to summarize the objectively existing innovation rules and methods researched and explored in behavioral innovation, and extract innovative elements, so as to effectively develop the innovation potential of ordinary people, enhance the innovation ability, so that ordinary people can also have innovative behaviors, engage in innovative activities and constantly improve the quality of the innovative activities, and carry out arranging and combining the objects that need innovation with innovative principles.

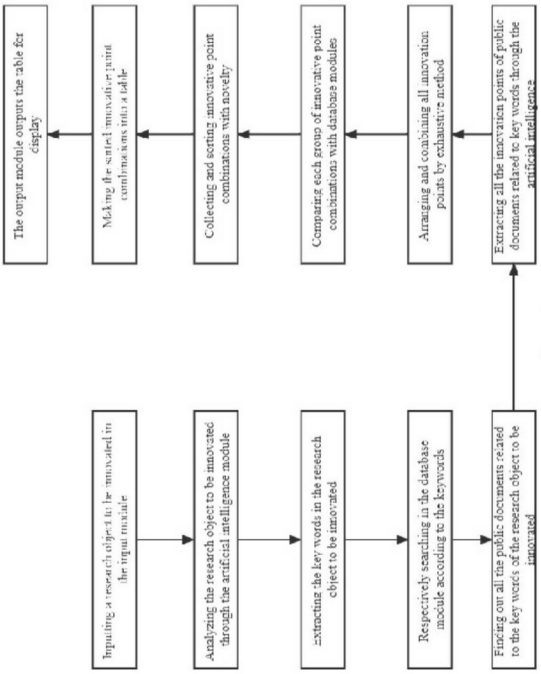

Figure 1

LU502887

Patent innovation method and system based on artificial intelligence

TECHNICAL FIELD

The invention relates to the technical field of intellectual property, and in particular to a patent innovation method and system based on artificial intelligence.

BACKGROUND

Most of the patents are inventions of parameter optimization, generally simple improvements to existing systems, and the improved technical principle has been applied in other fields or other patents.

It is often found that some people have many patents, while others don't. For those who have patents, the innovative methods used in their patents are mostly similar, and even many people apply for a large number of seemingly different patents on the same day, but they actually use the same technical principles. Many patent inventors just change the technical schemes used by predecessors to different scenes, and make new patents under new scenes, which leads to the lack of innovation and novelty of patent application documents, resulting in the low quality of patent application documents, which cannot meet the needs of society. In view of this, a patent innovation method and system based on artificial intelligence is proposed.

SUMMARY

The objective of the present invention is to provide a patent innovation method and system based on artificial intelligence, so as to solve the problem that the existing patent application documents are not innovative and novel enough in the background technology, which leads to the low quality of the patent application documents and cannot meet the needs of the society.

To achieve the above objectives, the present invention provides the following technical scheme.

A patent innovation method and system based on artificial intelligence, including the following modules:

an input module is used for inputting a research object to be innovated;

a database module is used to obtain patent technical documents and information published by countries all over the world;

an artificial intelligence module id used to analyse the research object to be innovated, extract key words in the research object to be innovated, then search in the database module according to the key words, find out the public documents related to the research object to be innovated, extract innovation points of all relevant documents through artificial intelligence, arrange and combine all the innovation points through an exhaustive method, and compare novelty with existing patents to select a new patent scheme of the research object to be innovated;

an output module is used to output the innovation results of artificial intelligence module, thus generating a large number of innovative patents.

Preferably, the data acquisition method of the data module includes the following steps:

S1, extracting innovation points: extracting individual literature innovation points from all patent technical documents in various countries, and collecting and sorting the innovation points;

S2, refining key words: refining the key words related to the innovative points collected and sorted in the patent literature, and collecting and sorting the key words; and

S3, storing in the database: collecting the patent documents, innovation points and key words, and storing the patent documents, innovation points and key words in the database module.

Preferably, the patent innovation method of artificial intelligence module includes the following steps:

S1, inputting a research object to be innovated in the input module;

S2, analysing the research object to be innovated through the artificial intelligence module, and segmenting the text of the research object to be innovated;

S3, extracting the key words in the research object to be innovated through the artificial intelligence module;

S4, respectively searching in the database module according to each keyword to find out all the public documents related to the key words of the research object to be innovated;

S5, extracting all the innovation points of public documents related to key words through the artificial intelligence;

S6, arranging and combining all innovation points by exhaustive method;

S7, comparing each group of innovative point combinations with database modules, and collecting and sorting innovative point combinations with novelty; and

S8: making the sorted innovative point combinations into a table, and the output module outputs the table for display.

Preferably, the database module is an open document database with the same technical field of the research object to be innovated.

Preferably, the database module is a public document database with the same key words as the research object to be innovated.

Preferably, the keyword in the research object to be innovated is selected as one keyword or multiple key words.

Preferably, the number of innovative point combinations with novelty after the innovative points are arranged and combined is represented by a mathematical formula: $S = 2^{n}-n-1-m(n>2,\ m>0)$, where S represents the total number of all innovative point permutations, n represents the number of innovative points of all related documents, and m represents the existing innovative point combinations in the database module.

Preferably, when n = 0, the number of innovation points of related literature is 0, that is, there is no related literature, indicating that the research object to be innovated is a new research object, and the patent applicants innovate independently;

when n = 1, the number of innovation points of related literatures is 1, that is, the number of related literatures is 1, indicating that the research object to be innovated is a new type of research object, and the patent applicants independently innovate;

when n = 2, m = 0, S = 1, the number of innovations in related literatures is 2, the number of existing innovative point combinations in database module is 0, and the innovative point combinations with novelty is 1; patent applicants may organize and construct the two innovations as new patent ideas;

when n = 2, m = 1, s = 0, the number of innovation points of related literatures is 2, the number of existing innovation point combinations in database module is 1, and the innovative point combinations with novelty is 0; patent applicants independently innovate.

Preferably, when S≥1, the patent applicant makes the sorted innovation point combination into a table through the output module, and outputs the table for display, and the innovation point combinations in the table are taken as new innovation ideas.

Compared with the prior art, the invention has the advantages that:

1. This patent innovation method and system based on artificial intelligence uses artificial intelligence to summarize the objective innovation laws and methods from the research and exploration of behavioural innovation, and extract innovative elements, so as to effectively develop the innovation potential of ordinary people, enhance the innovation ability, so that ordinary people can also have innovative behaviours, carry out innovative activities and constantly improve the quality of their innovative activities until the value of "innovation" is produced, and arrange and combine innovative objects that need innovation with innovative principles.

2. The patent innovation method and system based on artificial intelligence are suitable for unpopular or new facilities by searching multiple key words and related technical fields, and the number of comparative documents of research objects to be innovated is increased, so that the number of innovative point combinations is increased, and the research objects to be innovated are prevented from belonging to unpopular or new facilities, so that unpopular or new facilities may generate more new patent ideas, thus improving work efficiency.

3. The patent innovation method and system based on artificial intelligence are suitable for popular or well-known devices by setting a single keyword for searching. Because of the large number of related public documents of popular or well-known

devices, and there are enough innovations and a large number of combined innovations, thus improving the retrieval efficiency of documents and further improving the work efficiency of artificial intelligence.

BRIEF DESCRIPTION OF THE FIGURES

Fig. 1 is a flowchart of the patent innovation method of the artificial intelligence module of the present invention;

Fig. 2 is a flowchart of the data acquisition method of the data module of the present invention.

DESCRIPTION OF THE INVENTION

The technical solutions in the embodiments of the present invention will be clearly and completely described below with reference to the drawings in the embodiments of the present invention. Obviously, the described embodiments are only part of the embodiments of the present invention, but not all of them. Based on the embodiment of the present invention, all other embodiments obtained by ordinary technicians in the field without creative labour are within the scope of the present invention.

As shown in Figs. 1-2, the present invention provides the following two technical solutions.

Embodiment 1

A patent innovation method and system based on artificial intelligence. The patent innovation system includes the following modules:

an input module is used for inputting a research object to be innovated;

a database module is used to obtain patent technical documents and information published by countries all over the world;

an artificial intelligence module id used to analyse the research object to be innovated, extract key words in the research object to be innovated, then search in the database module according to the key words, find out the public documents related to

the research object to be innovated, extract innovation points of all relevant documents through artificial intelligence, arrange and combine all the innovation points through an exhaustive method, and compare novelty with existing patents to select a new patent scheme of the research object to be innovated;

an output module is used to output the innovation results of artificial intelligence module, thus generating a large number of innovative patents.

It is worth noting that, as shown in Fig. 2, the data acquisition method of the data module includes the following steps:

S1, extracting innovation points: extracting individual literature innovation points from patent technical documents of various countries, and collecting and sorting them out;

S2, refining key words: refining key words related to the innovation points collected and sorted out in patent literature, and collecting and sorting out them;

S3, storing in the database: binding and sorting the patent documents, innovations and key words, and storing them in the database module.

It is worth noting that, as shown in Figure 1, the patent innovation method of artificial intelligence module includes the following steps:

S1, extracting innovation points: extracting individual literature innovation points from all patent technical documents in various countries, and collecting and sorting the innovation points;

S2, refining key words: refining the key words related to the innovative points collected and sorted in the patent literature, and collecting and sorting the key words; and

S3, storing in the database: collecting the patent documents, innovation points and key words, and storing the patent documents, innovation points and key words in the database module.

Preferably, the patent innovation method of artificial intelligence module includes the following steps:

S1, inputting a research object to be innovated in the input module;

S2, analysing the research object to be innovated through the artificial intelligence module, and segmenting the text of the research object to be innovated;

S3, extracting the key words in the research object to be innovated through the artificial intelligence module;

S4, respectively searching in the database module according to each keyword to find out all the public documents related to the key words of the research object to be innovated;

S5, extracting all the innovation points of public documents related to key words through the artificial intelligence;

S6, arranging and combining all innovation points by exhaustive method;

S7, comparing each group of innovative point combinations with database modules, and collecting and sorting innovative point combinations with novelty; and

S8: making the sorted innovative point combinations into a table, and the output module outputs the table for display.

Furthermore, EXCEL in Microsoft OFFICE or Jinshan Table in Jinshan WPS can be used for the tables making in this embodiment.

It is worth noting that the database module in this embodiment is a public document database with the same key words as the research object to be innovated, and the database is larger.

In addition, the keyword in the research object to be innovated is selected as multiple key words, and the number of comparison files of the research object to be innovated is increased.

Furthermore, in this embodiment, the number of innovative point combinations with novelty after the innovative points are arranged and combined is represented by a mathematical formula: $S=2^{n}-n-1-m(n>2, m>0)$, where S represents the total number of all innovative point permutations, n represents the number of innovative points of all related documents, and m represents the existing innovative point combinations in the database module.

Preferably, when n = 0, the number of innovation points of related literature is 0, that is, there is no related literature, indicating that the research object to be innovated is a new research object, and the patent applicants innovate independently;

when n = 1, the number of innovation points of related literatures is 1, that is, the number of related literatures is 1, indicating that the research object to be innovated is a new type of research object, and the patent applicants independently innovate;

when n = 2, m = 0, S = 1, the number of innovations in related literatures is 2, the number of existing innovative point combinations in database module is 0, and the innovative point combinations with novelty is 1; patent applicants may organize and construct the two innovations as new patent ideas;

when n = 2, m = 1, s = 0, the number of innovation points of related literatures is 2, the number of existing innovation point combinations in database module is 1, and the innovative point combinations with novelty is 0; patent applicants independently innovate;

when S≥1, the patent applicant makes the sorted innovation point combination into a table through the output module, and outputs the table for display, and the innovation point combinations in the table are taken as new innovation ideas.

The embodiment is suitable for unpopular or new-type facilities, the number of comparison documents of research objects to be innovated is increased, the number of combinations of innovative points is increased, and the research objects to be innovated are prevented from belonging to unpopular or new-type facilities, so that unpopular or new-type facilities can also generate more new patent ideas, thereby improving work efficiency.

Embodiment 2

A patent innovation method and system based on artificial intelligence. The patent innovation system includes the following modules:

an input module is used for inputting a research object to be innovated;

a database module is used to obtain patent technical documents and information published by countries all over the world;

an artificial intelligence module id used to analyse the research object to be innovated, extract key words in the research object to be innovated, then search in the database module according to the key words, find out the public documents related to the research object to be innovated, extract innovation points of all relevant documents through artificial intelligence, arrange and combine all the innovation points through an exhaustive method, and compare novelty with existing patents to select a new patent scheme of the research object to be innovated;

an output module is used to output the innovation results of artificial intelligence module, thus generating a large number of innovative patents.

It is worth noting that, as shown in Fig. 2, the data acquisition method of the data module includes the following steps:

S1, extracting innovation points: extracting individual literature innovation points from patent technical documents of various countries, and collecting and sorting them out;

S2, refining key words: refining key words related to the innovation points collected and sorted out in patent literature, and collecting and sorting out them;

S3, storing in the database: binding and sorting the patent documents, innovations and key words, and storing them in the database module.

It is worth noting that, as shown in Figure 1, the patent innovation method of artificial intelligence module includes the following steps:

S1, extracting innovation points: extracting individual literature innovation points from all patent technical documents in various countries, and collecting and sorting the innovation points;

S2, refining key words: refining the key words related to the innovative points collected and sorted in the patent literature, and collecting and sorting the key words; and

S3, storing in the database: collecting the patent documents, innovation points and key words, and storing the patent documents, innovation points and key words in the database module.

Preferably, the patent innovation method of artificial intelligence module includes the following steps:

S1, inputting a research object to be innovated in the input module;

S2, analysing the research object to be innovated through the artificial intelligence module, and segmenting the text of the research object to be innovated;

S3, extracting the key words in the research object to be innovated through the artificial intelligence module;

S4, respectively searching in the database module according to each keyword to find out all the public documents related to the key words of the research object to be innovated;

S5, extracting all the innovation points of public documents related to key words through the artificial intelligence;

S6, arranging and combining all innovation points by exhaustive method;

S7, comparing each group of innovative point combinations with database modules, and collecting and sorting innovative point combinations with novelty; and

S8: making the sorted innovative point combinations into a table, and the output module outputs the table for display.

Furthermore, EXCEL in Microsoft OFFICE or Jinshan Table in Jinshan WPS can be used for table making in this embodiment.

It is worth noting that the database module in this embodiment is a public document database with the same key words as the research object to be innovated, thus reducing the number of comparison documents in the database.

In addition, the keyword in the research object to be innovated is selected as one keyword, which can also reduce the number of comparison files in the database.

Furthermore, in this embodiment, the number of innovative point combinations with novelty after the innovative points are arranged and combined is represented by a mathematical formula: $S=2^{n}-n-1-m(n>2, m>0)$, where S represents the total number of all innovative point permutations, n represents the number of innovative points of

all related documents, and m represents the existing innovative point combinations in the database module.

Preferably, when n = 0, the number of innovation points of related literature is 0, that is, there is no related literature, indicating that the research object to be innovated is a new research object, and the patent applicants innovate independently;

when n = 1, the number of innovation points of related literatures is 1, that is, the number of related literatures is 1, indicating that the research object to be innovated is a new type of research object, and the patent applicants independently innovate;

when n = 2, m = 0, S = 1, the number of innovations in related literatures is 2, the number of existing innovative point combinations in database module is 0, and the innovative point combinations with novelty is 1; patent applicants may organize and construct the two innovations as new patent ideas;

when n = 2, m = 1, s = 0, the number of innovation points of related literatures is 2, the number of existing innovation point combinations in database module is 1, and the innovative point combinations with novelty is 0; patent applicants independently innovate;

when S≥1, the patent applicant makes the sorted innovation point combination into a table through the output module, and outputs the table for display, and the innovation point combinations in the table are taken as new innovation ideas.

This embodiment is suitable for popular or well-known devices. Because of the large number of related public documents of popular or well-known devices, there are enough innovations and a large number of combined innovations, which improves the document retrieval efficiency and further improves the work efficiency of artificial intelligence.

The above shows and describes the basic principle, main features and advantages of the present invention. It should be understood by those skilled in the art that the present invention is not limited by the above-mentioned embodiments. The above-mentioned embodiments and descriptions are only preferred examples of the present invention, and are not intended to limit the present invention. Without departing from the spirit and scope of the present invention, the present invention will

undergo various changes and improvements, all of which fall within the scope of the claimed invention. The scope of that invention is define by the appended claim and their equivalents.

CLAIMS

1. A patent innovation method and system based on artificial intelligence, characterized by comprising the following modules:

an input module is used for inputting a research object to be innovated;

a database module is used to obtain patent technical documents and information published by countries all over the world;

an artificial intelligence module is used to analyse the research object to be innovated, extract key words in the research object to be innovated, then search in the database module according to the key words, find out the public documents related to the research object to be innovated, extract innovation points of all relevant documents through artificial intelligence, arrange and combine all the innovation points through an exhaustive method, and compare novelty with existing patents to select a new patent scheme of the research object to be innovated;

an output module is used to output the innovation results of artificial intelligence module, thus generating a large number of innovative patents.

2. The patent innovation method and system based on artificial intelligence according to claim 1, characterized in that the data acquisition method of the data module comprises the following steps:

S1, extracting innovation points: extracting individual literature innovation points from all patent technical documents in various countries, and collecting and sorting the innovation points;

S2, refining key words: refining the key words related to the innovative points collected and sorted in the patent literature, and collecting and sorting the key words; and

S3, storing in the database: collecting the patent documents, innovation points and key words, and storing the patent documents, innovation points and key words in the database module.

3. The patent innovation method and system based on artificial intelligence according to claim 1, characterized in that the patent innovation method of artificial intelligence module comprises the following steps:

S1, inputting a research object to be innovated in the input module;

S2, analysing the research object to be innovated through the artificial intelligence module, and segmenting the text of the research object to be innovated;

S3, extracting the key words in the research object to be innovated through the artificial intelligence module;

S4, respectively searching in the database module according to each keyword to find out all the public documents related to the key words of the research object to be innovated;

S5, extracting all the innovation points of public documents related to key words through the artificial intelligence;

S6, arranging and combining all innovation points by exhaustive method;

S7, comparing each group of innovative point combinations with database modules, and collecting and sorting innovative point combinations with novelty; and

S8: making the sorted innovative point combinations into a table, and the output module outputs the table for display.

4. The patent innovation method and system based on artificial intelligence according to claim 3, characterized in that the database module is an open document database with the same technical field of the research object to be innovated.

5. The patent innovation method and system based on artificial intelligence according to claim 3, characterized in that the database module is a public document database with the same key words as the research object to be innovated.

6. The patent innovation method and system based on artificial intelligence according to claim 3, characterized in that the keyword in the research object to be innovated is selected as one keyword or multiple key words.

7. The patent innovation method and system based on artificial intelligence according to claim 3, characterized in that the number of innovative point combinations with novelty after the innovative points are arranged and combined is represented by a mathematical formula: $S=2^{n}-n-1-m(n>2, m>0)$, where S represents the total number of all innovative point permutations, n represents the number of

innovative points of all related documents, and m represents the existing innovative point combinations in the database module.

8. The patent innovation method and system based on artificial intelligence according to claim 7, characterized in that when n = 0, the number of innovation points of related literature is 0, that is, there is no related literature, indicating that the research object to be innovated is a new research object, and the patent applicants innovate independently;

when n = 1, the number of innovation points of related literatures is 1, that is, the number of related literatures is 1, indicating that the research object to be innovated is a new type of research object, and the patent applicants independently innovate;

when n = 2, m = 0, S = 1, the number of innovations in related literatures is 2, the number of existing innovative point combinations in database module is 0, and the innovative point combinations with novelty is 1; patent applicants may organize and construct the two innovations as new patent ideas;

when n = 2, m = 1, s = 0, the number of innovation points of related literatures is 2, the number of existing innovation point combinations in database module is 1, and the innovative point combinations with novelty is 0; patent applicants independently innovate.

9. The patent innovation method and system based on artificial intelligence according to claim 7, characterized in that: when S≥1, the patent applicant makes the sorted innovation point combination into a table through the output module, and outputs the table for display, and the innovation point combinations in the table are taken as new innovation ideas.

Revendications

1. Un procédé et un système d'innovation de brevet basés sur l'intelligence artificielle, caractérisés en ce qu'ils comprennent les modules suivants:

un module d'entrée est utilisé pour entrer un objet de recherche à innover;

un module de base de données est utilisé pour obtenir des documents techniques et des informations en matière de brevets publiés par des pays du monde entier;

un module d'intelligence artificielle est utilisé pour analyser l'objet de recherche à innover, extraire des mots clés dans l'objet de recherche à innover, rechercher ensuite dans le module de base de données en fonction des mots clés, trouver les documents publics relatifs à l'objet de recherche à innover, extraire les points d'innovation de tous les documents pertinents par l'intelligence artificielle, organiser et combiner tous les points d'innovation par une méthode exhaustive, et comparer la nouveauté avec les brevets existants pour sélectionner un nouveau système de brevets de l'objet de recherche à innover;

un module de sortie est utilisé pour sortir les résultats d'innovation d'un module d'intelligence artificielle, générant ainsi un grand nombre de brevets innovants.

2. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 1, caractérisés en ce que le procédé d'acquisition de données du module de données comprend les étapes suivantes:

S1, extraire les points d'innovation: extraire des points d'innovation de la littérature individuelle de tous les documents techniques relatifs aux brevets dans divers pays, et recueillir et trier les points d'innovation;

S2, affiner les mots clés: affiner les mots clés relatifs aux points innovants collectés et triés dans la littérature en matière de brevets, et collecter et trier les mots clés; et

S3, stocker dans la base de données: collecter les documents de brevet, les points d'innovation et les mots clés, et stocker les documents de brevet, les points d'innovation et les mots clés dans le module de base de données.

3. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 1, caractérisés en ce que le module d'intelligence artificielle de la méthode d'innovation de brevet comprend les étapes suivantes:

S1, introduire un objet de recherche à innover dans le module d'entrée;

S2, analyser l'objet de recherche à innover par le biais du module d'intelligence artificielle, et segmenter le texte de l'objet de recherche à innover;

S3, extraire les mots clés de l'objet de recherche à innover par le biais du module d'intelligence artificielle;

S4, rechercher respectivement dans le module de base de données, en fonction de chaque mot clé, tous les documents publics relatifs aux mots clés de l'objet de recherche à innover;

S5, extraire tous les points d'innovation des documents publics liés aux mots clés par l'intelligence artificielle;

S6, organiser et combiner tous les points d'innovation par une méthode exhaustive;

S7, comparer chaque groupe de combinaisons de points innovantes avec des modules de base de données, et collecter et trier des combinaisons de points innovantes avec la nouveauté; et

S8, transformer les combinaisons de points innovantes triées en une table, et le module de sortie sort la table pour affichage.

4. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 3, caractérisés en ce que le module base de données est une base de données de documents ouverte avec le même domaine technique de l'objet de recherche à innover.

5. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 3, caractérisés en ce que le module base de données est une base de données de documents publics avec les mêmes mots clés que l'objet de recherche à innover.

6. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 3, caractérisés en ce que le mot-clé dans l'objet de recherche à innover est sélectionné comme un mot-clé ou plusieurs mots-clés.

7. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 3, caractérisés en ce que le nombre de combinaisons de points innovantes avec la nouveauté après que les points innovants sont arrangés et combinés est représenté par une formule mathématique: $S=2^{n}-n-1-m(n>2, m>0)$, S représente le nombre total de toutes les permutations de points innovantes, n représente le nombre de points innovants de tous les documents connexes, et m représente les combinaisons de points innovantes existantes dans le module de base de données.

8. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 7, caractérisés en ce que lorsque n = 0, le nombre de points d'innovation de la littérature connexe est de 0, c'est-à-dire qu'il n'existe pas de documentation connexe, indiquant que l'objet de recherche à innover est un nouvel objet de recherche, et les demandeurs de brevet innovent de manière indépendante;

lorsque n = 1, le nombre de points d'innovation des littératures connexes est de 1, c'est-à-dire que le nombre de publications connexes est de 1, indiquant que l'objet de recherche à innover est un nouveau type d'objet de recherche, et les demandeurs de brevet innovent de manière indépendante;

lorsque n = 2, m = 0, S = 1, le nombre d'innovations dans les littératures connexes est de 2, le nombre de combinaisons de points innovantes existantes dans le module de base de données est de 0, et les combinaisons de points innovantes avec nouveauté est de 1; les déposants de brevets peuvent organiser et construire les deux innovations en tant que nouvelles idées en matière de brevets;

lorsque n = 2, m = 1, s = 0, le nombre de points d'innovation des littératures connexes est de 2, le nombre de combinaisons de points d'innovation existantes dans

le module de base de données est de 1, et les combinaisons de points innovantes avec nouveauté est de 0; les déposants de brevets innovent indépendamment.

9. Le procédé et un système d'innovation de brevet basés sur l'intelligence artificielle selon la revendication 7, caractérisés en ce que: lorsque S ≥ 1, le demandeur de brevet transforme la combinaison triée de points d'innovation en un tableau par l'intermédiaire du module de sortie, et génère le tableau pour affichage, et les combinaisons de points d'innovation dans le tableau sont considérées comme de nouvelles idées d'innovation.

FIGURES

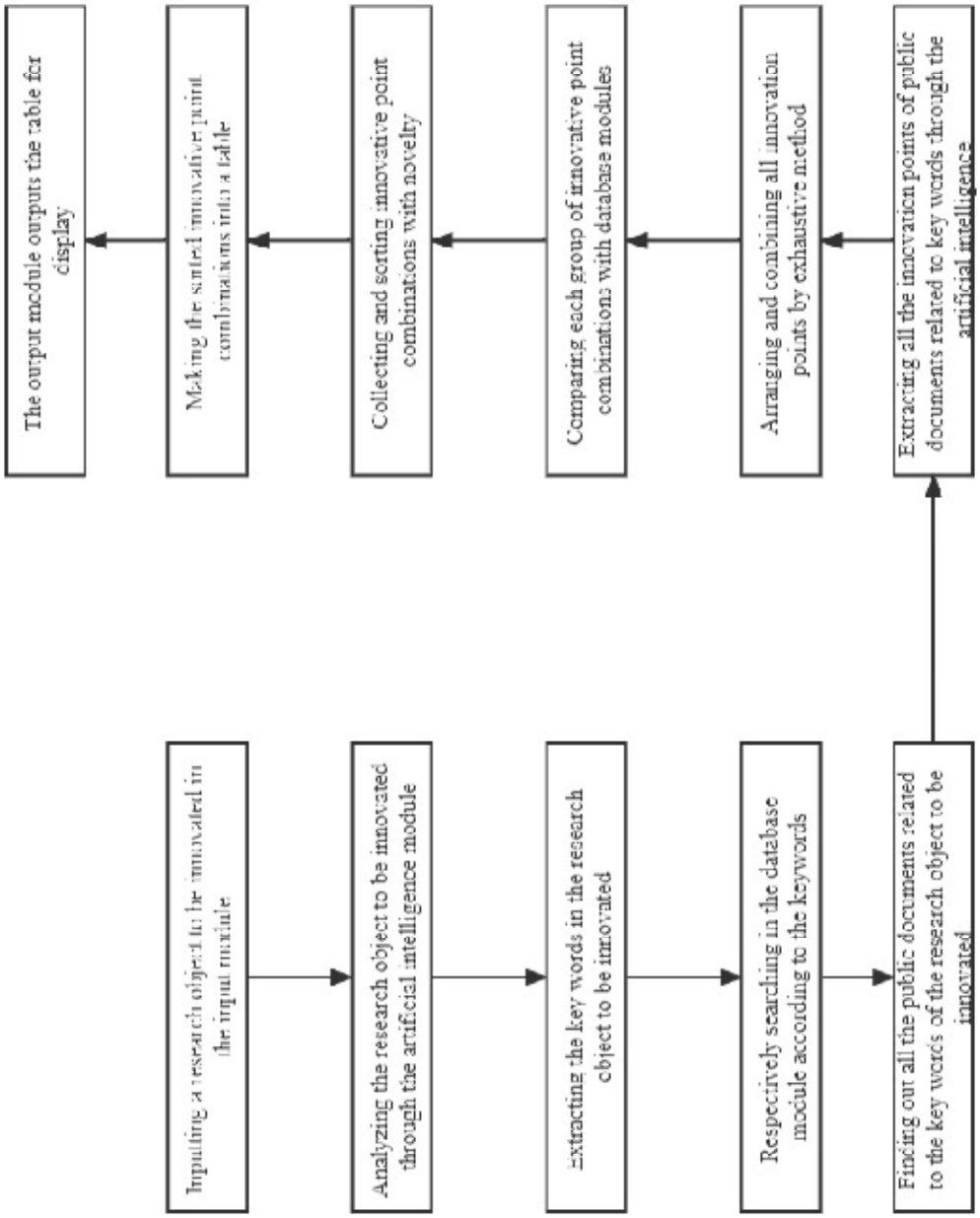

Figure 1

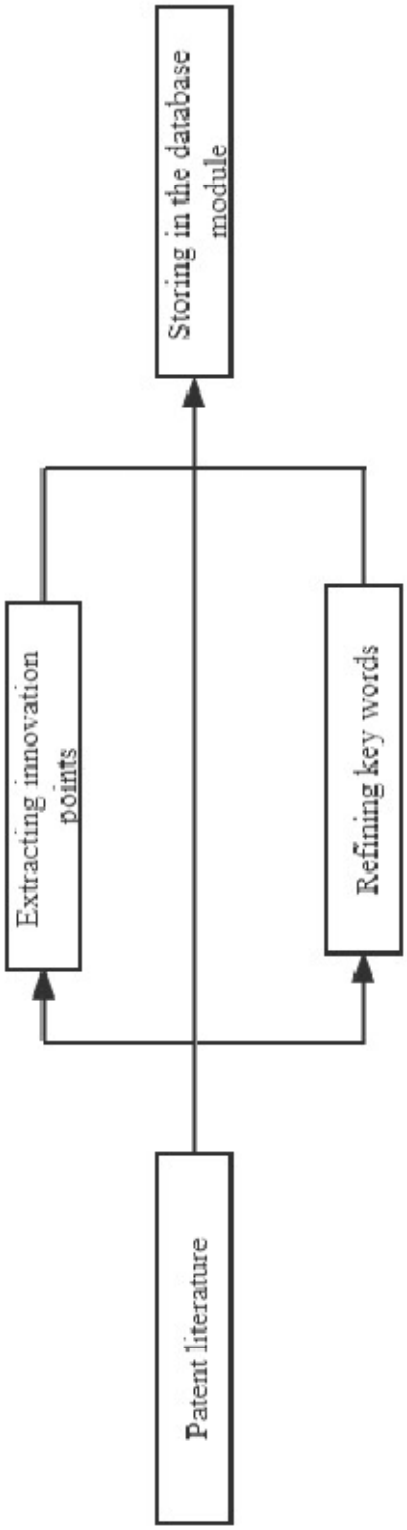
Patent literature
Extracting innovation points
Refining key words
Storing in the database module

Figure 2